De Mathematics Competitions

2nd Annual

DMC 12 A

Wednesday, September 30, 2026



INSTRUCTIONS

1. DO NOT TURN TO THE NEXT PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple-choice competition. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 6 points for each correct answer, 1.5 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, or computing devices are allowed. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. Before beginning the competition, your competition manager will ask you to record your name on the answer sheet.
8. You will have 75 minutes to complete the competition once your competition manager tells you to begin.
9. When you finish with the competition, sign your name in the space provided on the answer sheet and complete the demographic information questions on the back of the answer sheet.

The problems and solutions for this DMC 12 were prepared
by the DMC Committee under the direction of:

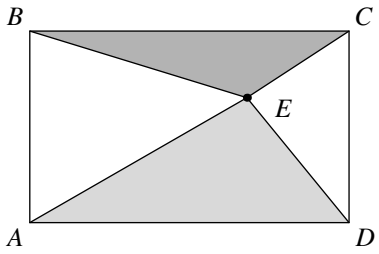
Taiki Aiba and Ryan Shin

The DMC Committee reserves the right to disqualify scores from a school if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Students who score well on this DMC 12 will be invited to take the De Invitational Mathematics Examination (DIME), which is already out and is due on Friday, January 29, 2027. More details about the DIME can be found on our site.

1. Point E lies inside rectangle $ABCD$, as shown below. The area of $ABCD$ is 46, and the area of $\triangle AED$ is 15. What is the area of $\triangle BEC$?



(A) $\frac{15}{2}$ **(B)** 8 **(C)** $\frac{31}{3}$ **(D)** $\frac{23}{2}$ **(E)** 15

2. At Test Academy, 50% of the students had eggs for breakfast, 60% had pancakes, and 20 students had both. Every student had at least one of the two. How many students attend Test Academy?

(A) 100 **(B)** 160 **(C)** 200 **(D)** 240 **(E)** 400

3. What fraction of the sum $1 + 2 + 3 + \cdots + 49$ comes from the terms that are multiples of 7?

(A) $\frac{1}{7}$ **(B)** $\frac{4}{25}$ **(C)** $\frac{8}{49}$ **(D)** $\frac{1}{6}$ **(E)** $\frac{1}{5}$

4. A list of positive integers begins with 20^{26} and ends with 1. Each term after the first is a divisor of the term before it and is smaller than that term. What is the greatest possible number of terms in the list?

(A) 27 **(B)** 53 **(C)** 78 **(D)** 79 **(E)** 80

5. In $\triangle ABC$, $\angle B = 60°$, $AC = 4$, and $AB^3 + BC^3 = 112$. What is the perimeter of $\triangle ABC$?

(A) 10 **(B)** 11 **(C)** 12 **(D)** 13 **(E)** 15

6. Five positive integers consist of two equal even numbers and three (not necessarily distinct) odd numbers. Some four of them have sum 125, some three of them have sum 60, and some two of them have sum 64. What is the sum of all five numbers?

(A) 124 **(B)** 157 **(C)** 181 **(D)** 189 **(E)** 249

7. What is the sum of all positive real numbers x such that

$$\frac{(x^3)^{\log_{16} x}}{x^{\log_{16} x}} = x?$$

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4 **(E)** 5

8. Let $P(x)$ be a quadratic polynomial with leading coefficient 1 such that $P(1) = 4$ and

$$P(x) + P\left(\frac{1}{x}\right) = P\left(x + \frac{1}{x}\right)$$

for all nonzero real x. What is $P(5)$?

(A) 32 **(B)** 35 **(C)** 37 **(D)** 40 **(E)** 42

9. Circles ω_1 and ω_2, each with radius 3, are internally tangent to a circle Ω with radius 9. Their points of tangency are chosen independently and uniformly at random on Ω. What is the probability that ω_1 and ω_2 intersect?

(A) $\frac{1}{6}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{3}$ **(D)** $\frac{1}{2}$ **(E)** $\frac{2}{3}$

10. Chloe has a 5×7 sheet of paper. Each move, Owen chooses one piece and tears it into two pieces of equal area. After some number of moves, exactly 5 pieces have area less than 1, and every other piece has area greater than 2. What is the least possible number of moves?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 12

11. The four roots of $x^4 - ax^2 - 1$, where a is a real number, are the vertices of a quadrilateral in the complex plane with perimeter 12. What is a^2?

(A) 77 **(B)** 79 **(C)** 81 **(D)** 83 **(E)** 85

12. Isabella draws a card at random from five cards numbered 0, 1, 2, 3, 4 and stands at that point on a number line. Each second, she steps 1 unit left or right with equal probability, except that once she is at 0 or 4 (including at the start), she stays there forever. What is the probability that after 10 seconds she is standing on an even number?

(A) $\frac{4}{5}$ **(B)** $\frac{9}{10}$ **(C)** $\frac{19}{20}$ **(D)** $\frac{39}{40}$ **(E)** $\frac{79}{80}$

13. Let N be the number of nonzero coefficients in the expansion of

$$\left(x + 2x^2 + 3x^3 + \cdots + 2026x^{2026}\right)^2$$

that are even. What is the sum of the digits of N?

(A) 5 **(B)** 7 **(C)** 13 **(D)** 14 **(E)** 15

14. Ken labels each vertex of a regular pentagon with a nonzero digit, not necessarily different, and draws the pentagon's five diagonals, which cross at five points. Each crossing point is labeled with the product of the labels at the four endpoints of the two diagonals through it. The sum of the five crossing labels is three times the product of the five vertex labels. Labelings that can be obtained from one another by a rotation and/or reflection are considered different. How many labelings of the vertices are possible?

(A) 75 **(B)** 95 **(C)** 100 **(D)** 105 **(E)** 120

15. In $\triangle ABC$, $\angle ABC = 26^\circ$. Point D lies on $\overline{BC}$ so that $\overline{AD}$ bisects $\angle BAC$, and $AD > CD$. What is the greatest possible integer value of the degree measure of $\angle ADC$?

(A) 52 **(B)** 68 **(C)** 77 **(D)** 78 **(E)** 102

16. On a 5×5 chessboard, a black king is in one corner and a white king is in the diagonally opposite corner. A bishop attacks a square on one of its diagonals exactly when no piece stands strictly between them. In how many ways can a black bishop and a white bishop be placed on two other squares so that neither bishop can attack the king of the opposite color?

(A) 375 **(B)** 380 **(C)** 383 **(D)** 386 **(E)** 506

17. In regular hexagon $ABCDEF$ with side length 7, lines BC and DE meet at G. Point P is chosen uniformly at random on the perimeter of pentagon $ABGEF$. What is the expected area of the convex quadrilateral with vertices A, C, D, and P?

(A) $\frac{49\sqrt{3}}{2}$ **(B)** $30\sqrt{3}$ **(C)** $33\sqrt{3}$ **(D)** $35\sqrt{3}$ **(E)** $\frac{196\sqrt{3}}{5}$

18. A real number x satisfies

$$(\cos x + 20 \sec x)^2 + (\sin x + 26 \csc x)^2 = 3050.$$

What is $|20 \tan x - 26 \cot x|$?

(A) 23 **(B)** 25 **(C)** 27 **(D)** 29 **(E)** 31

19. In $\triangle ABC$ with $\angle B = 90^\circ$ and $AB = BC = 2\sqrt{2}$, circle ω passes through the midpoints of $\overline{AB}$, $\overline{BC}$, and $\overline{CA}$. Points D and E lie on ω with $\angle DBE = 90^\circ$ and $\angle BDE = 60^\circ$, and D is closer to A than E is. What is the area of non-convex pentagon $ACEBD$?

(A) $2 + \frac{\sqrt{3}}{2}$ **(B)** 3 **(C)** $\frac{5+\sqrt{3}}{2}$ **(D)** $2 + \sqrt{3}$ **(E)** $2 + 2\sqrt{3}$

20. An ellipse has foci A and B with $AB = 2$. A laser fired from A hits the ellipse at P, reflects off the ellipse, and hits the ellipse again at Q, with $AP = 2$ and $AQ = 3$. What is BP?

(A) $\frac{1+\sqrt{41}}{4}$ **(B)** $\frac{3+\sqrt{41}}{4}$ **(C)** $\frac{5+\sqrt{41}}{4}$ **(D)** $\frac{11+\sqrt{41}}{4}$ **(E)** $\frac{3+\sqrt{41}}{2}$

21. Let $\lfloor r \rfloor$ denote the greatest integer less than or equal to r. Let S be the region consisting of all points (x, y) with $0 \le x \le 1$ and $0 \le y \le 1$ for which there exist ordered pairs (a, b) and (c, d) of nonnegative real numbers, neither equal to $(0, 0)$, such that

$$\left\lfloor \frac{2025ax^2 + 2025by^2}{2026a + b} \right\rfloor \neq \left\lfloor \frac{2025cx^2 + 2025dy^2}{2026c + d} \right\rfloor.$$

What is the area of S?

(A) $\frac{17}{18}$ **(B)** $\frac{26}{27}$ **(C)** $\frac{29}{30}$ **(D)** $\frac{131}{135}$ **(E)** $\frac{44}{45}$

22. Let positive integers $a_1, a_2, a_3, \ldots$ satisfy $a_{n+2} = a_{n+1} + a_n$ for all $n \ge 1$. Suppose that one of the integers equals 2026. What is the least possible value of a_3?

(A) 24 **(B)** 46 **(C)** 68 **(D)** 80 **(E)** 92

23. For a real number a and a nonnegative integer k, define

$$\binom{a}{k} = \frac{a(a-1)(a-2)\cdots(a-k+1)}{k!},$$

with $\binom{a}{0} = 1$. What is the value of the sum

$$\sum_{j=0}^{\infty} \binom{1/2}{3j}?$$

(A) $\frac{2+\sqrt{2}+\sqrt{3}}{5}$ **(B)** $\frac{1+\sqrt{2}+\sqrt{3}}{4}$ **(C)** $\frac{\sqrt{2}+\sqrt{3}}{3}$ **(D)** $\frac{\sqrt{10}}{3}$ **(E)** $\frac{1+\sqrt{5}}{3}$

24. A sequence of positive integers $a_1, a_2, a_3, \ldots, a_{11}, b_1, b_2, b_3, \ldots, b_{11}$ satisfies $a_1 = b_1 = 1$ and $a_{11} = b_{11} = 27$. For each $2 \le i \le 11$, $a_i - a_{i-1}$ is a positive even integer, and $b_i - b_{i-1}$ is a positive odd integer. Also, $a_k = b_k$ for every odd k. How many such sequences are there?

(A) 1440 **(B)** 10,880 **(C)** 20,160 **(D)** 21,760 **(E)** 43,520

25. Convex pentagon $ABCDE$ has $CD = DE = 3$, $\overline{BC} \perp \overline{DE}$, $\angle BAC = \angle CAD$, and

$$\angle ABC = \angle AEC = \angle ACD = 90^\circ.$$

What is the area of $ABCDE$?

(A) $17\sqrt{2}$ **(B)** $18\sqrt{2}$ **(C)** $20\sqrt{2}$ **(D)** $23\sqrt{2}$ **(E)** $25\sqrt{2}$